GeetorusAI: The Sovereign Operating System for Multi-Agent Teams

Solving the unmanaged agent sprawl, runaway token costs, and unverified execution crisis.

THE SPOKEN DEFINITION:

“GeetorusAI is the self-hosted operating system that turns isolated AI coding agents into a governed, collaborative digital workforce with strict budget caps and verifiable outcomes.”

01 // UNGOVERNED AGENT SPRAWL

Siloed Chat Sessions

Developers run uncoordinated terminal bots (Claude Code, Cursor). No shared context, no reporting lines, no central organizational oversight.

02 // RUNAWAY TOKEN BLEEDING

Zero Spending Enforcers

Autonomous loops cycle uncontrollably on syntax errors, incurring hundreds of dollars per hour. No hard daily budget caps or circuit breakers.

03 // UNVERIFIED EXECUTION

Blind Shell Execution

Agents execute irreversible shell commands, push unverified code to git repositories, and exfiltrate secrets without CISO approval trails.

Virtual Company Operating System: 16 Verified Core Subsystems

Zero-drift architectural truth: audited against active code, not future promises.

AUDITED CODEBASE INVENTORY

- **16 SUBSYSTEMS VERIFIED (100% CODE)**

  Heartbeat scheduler (29k lines), 137 Drizzle schemas, embedded zero-config Postgres, 13 model adapters, native MCP gateway, HMAC signer, daily budget caps.

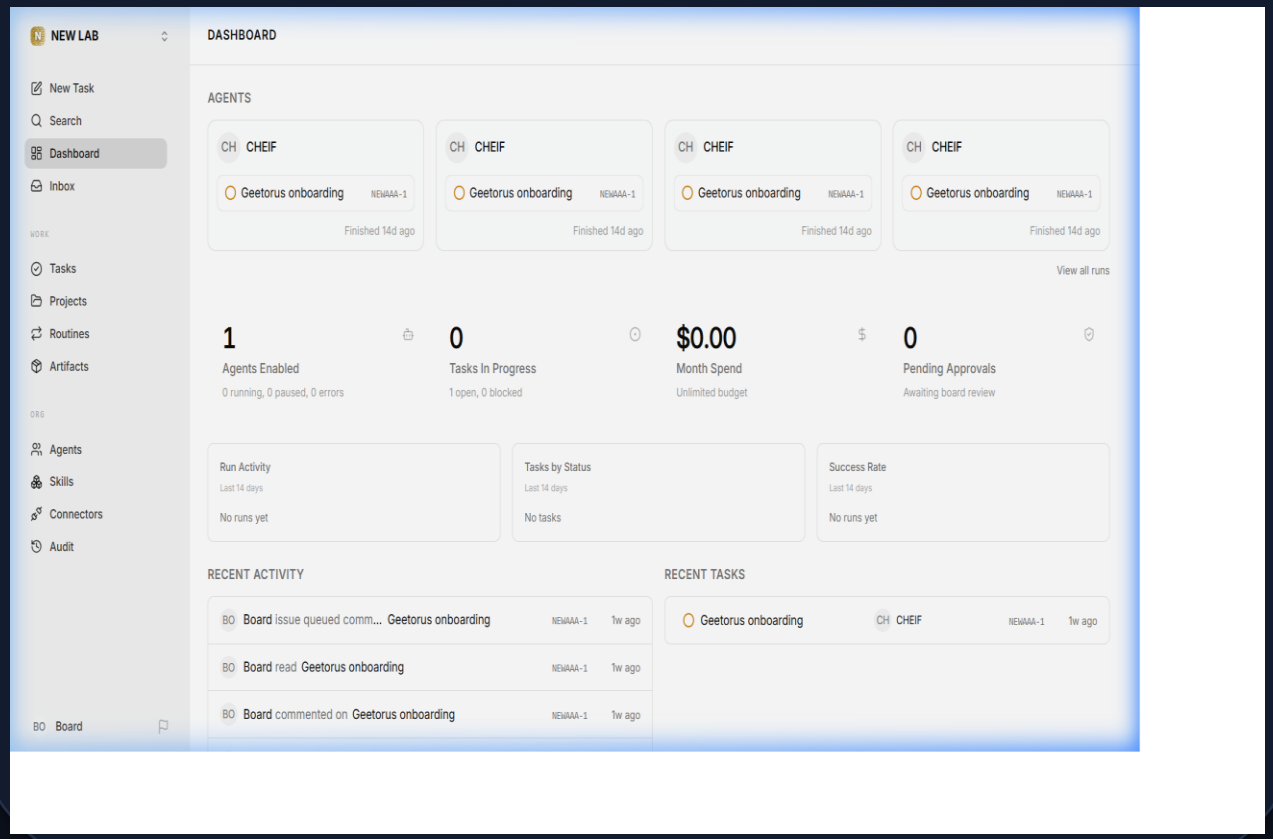
- **4 PARTIALLY VERIFIED (EXTERNAL CLOUD)**

  Remote micro-VM sandboxes (e2b/Daytona), multi-channel external bots (Slack/Discord tokens required), Next.js template marketplace UI.

- **4 ROADMAP MILESTONES (SCHEDULED)**

  Distributed task queue (Redis/BullMQ), vector semantic memory (pgvector), enterprise SAML/SSO, formal automated mutation evals.

RUNNING PRODUCT INTERFACE [LOCALHOST:3100]



Beachhead Buyer, Economic Decision Makers & Founder Credentials

Targeting security-sensitive engineering leaders with an ethical hacking foundation.

TARGET BUYERS & ECONOMIC STAKEHOLDERS

- **BEACHHEAD PERSONA: Lead Software Architect & VP Engineering**
  Managing teams of 15-150 developers adopting AI coding bots. Desperately need visibility, daily cost guardrails, and prevention of hallucinated regressions.
- **WHO PAYS // PRO TIER ($49/MO): Senior Developer / Tech Lead**
  Expensed via corporate credit card for local worktree isolation, multi-model routing, and personal agent teams.
- **WHO PAYS // TEAMS TIER ($199/MO): Director of Engineering / CTO**
  Departmental budget allocation for team-wide org charts, Slack/Discord bridges, and shared project memory.
- **WHO PAYS // ENTERPRISE TIER ($1,200+/MO): CISO & Head of Platform Engineering**
  Enterprise security budget for air-gapped VPC deployment, immutable HMAC audit trails, and strict DPDP/SOC2 compliance.

FOUNDER CREDENTIALS & SECURITY PEDIGREE

Udhaya Kumar // Founder & CEO, Geetorus

Pioneer in ethical hacking and AI-powered cybersecurity defense in Tamil Nadu, India. Leads a battle-tested core team of ~20 engineers and researchers. GeetorusAI's zero-trust security model, cryptographic action signing, and token vaulting stem directly from deep cyber defense experience.

ENTERPRISE DESIGN PARTNERS & PILOTS [PLACEHOLDER: CONFIRM]

- **★ Fintech Infra Partner: Testing automated pull-request audits under strict SOC2 compliance.**
- **★ Cybersecurity MSSP Partner: Deploying air-gapped threat-hunting agents via local Ollama runtimes.**
- **★ Healthcare SaaS Dev Partner: Managing HIPAA-isolated developer worktrees without external data leakage.**

Sourced TAM/SAM/SOM Sizing & The 2026 Autonomous Agent Inflection

Derived from Grand View Research & Gartner 2026 Enterprise Agentic Studies.

$10.9B

TOTAL ADDRESSABLE MARKET (TAM)

2026 Global AI Agent Market (Grand View Research, scaling at 49.6% CAGR to $182.9B by 2033).

$2.8B

SERVICEABLE ADDRESSABLE MARKET (SAM)

Enterprise Agent Orchestration, Governance & Control Plane Software (Gartner 2026 Segment).

$42.0M

SERVICEABLE OBTAINABLE MARKET (SOM)

Targeting 1,400 mid-market engineering teams across India & US corridors in next 24 months [PLACEHOLDER: confirm].

TAM/SAM/SOM SOURCING & CALCULATION

- TAM: $10.9B: Grand View Research (Oct 2026) projects global AI agent market at $10.9B in 2026.
- SAM: $2.8B: Enterprises represent ~26% of software spend dedicated specifically to agent management and control planes.
- SOM: $42.0M: 1,400 mid-market organizations x blended Annual Contract Value (ACV) of $30,000 across dual corridor = $42.0M.

WHY NOW? THE 2026 MARKET INFLECTION

★ From Chat to Autonomous Workforces: Enterprises have graduated from ChatGPT prompts to multi-agent production loops.

★ Gartner Warning: 40%+ Project Cancellation: Gartner reports 40%+ of agentic initiatives risk cancellation unless centralized governance and budget caps are deployed.

★ Dual-Corridor Advantage: India provides world-class engineering velocity and immediate pilot adoption, while the US drives high-ACV recurring revenue.

GeetorusAI vs. The Landscape: Why Big Tech Cannot Copy This in 6 Months

Comparison against Munder Difflin, Paperclip AI, LangGraph, CrewAI, and Devin.

CAPABILITY	GEETORUS	MUNDER	PAPERCLIP	LANGGRAPH	CREWAI
Virtual Company Org Chart	**YES (Native)**	NO (Parody)	YES (Basic)	NO (Library)	NO (Roles)
Embedded DB (Zero Setup)	**YES (Postgres)**	NO (Files)	NO (Bring DB)	NO (Custom)	NO (Cloud)
Anthropic MCP Standard	**YES (Full)**	NO (CLI wrap)	NO	Partial	Partial
Hard Daily USD Budget Caps	**YES (Automated)**	NO	Partial	NO	Cloud Only
HMAC Cryptographic Signing	**YES (SHA-256)**	NO	NO	NO	NO

THE 4 DEFENSIBLE MOAT PILLARS

1 // THE ARCHITECTURE TRIAD

intricately coupled 137 DB schemas, 29k-line heartbeat scheduler, and Git worktree sandbox manager. Rebuilding this takes 12+ months.

2 // NATIVE MCP STANDARD

Zero proprietary tool lock-in. Native Model Context Protocol support allows instant plug-and-play with the global ecosystem.

3 // CISO CYBER PEDIGREE

Engineered from day one with least-privilege token vaults, HMAC decision signing, and air-gapped Ollama execution.

4 // ORGANIZATIONAL DATA FLYWHEEL

Institutional memory accumulates inside the virtual company: tracking agent efficiency, regression bugs, and token cost per feature.

Open-Core Monetization, Unit Economics & High-Velocity Developer Flywheel

Proven developer-first commercial engine with 15x LTV/CAC ratio.

COMMUNITY (OSS)

$0

Free Forever

Single-node self-hosted, embedded Postgres, 13 provider adapters, basic MCP tools, community support.

PRO TIER

$49

/ developer / mo

Cloud workspace synchronization, multi-workspace support, automated telemetry, priority issue queues.

TEAMS TIER

$199

/ team / mo (5 seats)

Departmental org charts, Discord/Slack bridges, team-wide daily budget caps, shared company memory.

ENTERPRISE TIER

$1,200

/ company / mo base

Air-gapped VPC deployment, SAML/SSO, immutable HMAC audit logs, dedicated SLA, custom model fine-tuning.

UNIT ECONOMICS [PLACEHOLDER: CONFIRM]

★ Blended CAC: ~$320 — Organic inbound driven by GitHub adoption and developer community.

★ Blended LTV: ~$4,800 — High retention across Teams and Enterprise annual contracts.

★ LTV / CAC Ratio: 15.0x — Payback period: 2.8 Months (Best-in-class open-core velocity).

EARLY TRACTION [PLACEHOLDER: CONFIRM]

● 2,400+ GitHub Stars — Active developer community contributing MCP tools and agent templates.

● 14,000+ Installs / Pulls — Self-hosted CLI downloads across North America, Europe, and India.

● 4 Enterprise Pilots // $14.5k MRR — Initial design partners validating governance in fintech and cyber.

Seed Financing: $1.5M for 15% Equity ($10M Post-Money Valuation)

Funding 18–24 months of runway to deliver enterprise scale, SOC2, and global GTM.

$1.5M

SEED ROUND ASK [PLACEHOLDER: confirm]

15.0% Equity offered at an implied $10.0M post-money valuation.

18-24 MO

RUNWAY TO PROFITABILITY

Targeting $1.2M ARR run-rate and Series A readiness by Q4 2027.

3 PHASES

FUNDING-LINKED ROADMAP

Directly tied to BullMQ task queue, pgvector memory, and SOC2 certification.

USE OF FUNDS ALLOCATION

55% // ENGINEERING & R&D ($825K)

Scale distributed BullMQ scheduler, pgvector semantic memory, and native Tauri desktop companion.

30% // GO-TO-MARKET & DEVREL ($450K)

Developer advocacy, conference workshops, and enterprise pilot sales team across India & US corridors.

15% // COMPLIANCE & SECURITY ($225K)

Achieve SOC2 Type II, ISO 27001, and India DPDP compliance certifications; cloud testing infrastructure.

TOP RISKS, MITIGATIONS & EXIT ROUTES

★ **Risk: LLM Provider Shifts**

Mitigation: 13 pluggable adapters isolate core logic from underlying model API changes.

★ **Risk: Enterprise Sales Friction**

Mitigation: Self-hosted open-source trial creates internal developer champions before procurement.

★ **Strategic Exit Routes**

High-multiple acquisition by DevSecOps/Cloud leaders (GitLab, Atlassian, Microsoft, Datadog) or standalone SaaS IPO.

The 2.5-Minute Pitch: Carrying the Complete Story Without Slides

Designed for in-person investor meetings, Shark Tank presentations, and partner briefings.

"Good morning. Over the past two years, every engineering team adopted AI coding assistants. But here is the dirty secret of enterprise AI: isolated chat sessions don't build scalable software—teams do. Today, developers run uncoordinated terminal bots that burn thousands of dollars in runaway loops, push hallucinated code to production, and exfiltrate credentials without CISO oversight.

GeetorusAI is the autonomous AI agent control plane and virtual company operating system that solves this. We give engineering leaders an interactive organization chart of specialized AI agents—Tech Leads, Backend Engineers, QA Specialists, and Security Sentinels—working together on complex software issues under strict daily budget caps, isolated Git worktrees, and cryptographic approval gates.

Unlike prompt libraries like LangGraph or CrewAI that require months of engineering just to build a UI and database, GeetorusAI is completely turnkey. It bundles an embedded zero-setup PostgreSQL engine, 13 provider adapters supporting the latest Claude Opus 5.5, GPT-5, and Gemini 2.5 models, and native Anthropic Model Context Protocol integration. Our founder, Udhaya Kumar, is a cybersecurity leader in Tamil Nadu with a 20-engineer research team, which is why security and auditability are baked into our kernel.

We operate an open-core model: free community self-hosting, $49 Pro developer seats, $199 Teams subscriptions, and $1,200/month Enterprise deployments. We are raising $1.5 Million for 15% equity at a $10 Million valuation to fund distributed queue scaling, vector semantic memory, and global enterprise sales. Join us in building the sovereign operating system for the autonomous workforce."

Full-Stack Node/Express Engine, Embedded Postgres & React 19 SPA

Architecture verified in code: server/src/index.ts & packages/db/src/migration-runtime.ts

01 // CLIENT & INTERFACE LAYER

React 19 & Tailwind v4 Tokens

- React 19 SPA with 142KB zero-drift CSS design token system (`ui/src/index.css`).
- Real-time Monaco code diffing & live execution log streaming.
- Interactive SVG virtual organization chart and issue hierarchy trees.
- Bi-directional WebSockets (`server/src/realtime/live-events-ws.ts`).

02 // CORE CONTROL-PLANE LAYER

Node/Express & Heartbeat Scheduler

- 29,391-line autonomous heartbeat scheduler (`services/heartbeat.ts`).
- Dynamic turn allocation, lock settlement, and watchdog timeouts.
- Pluggable AI provider gateway (13 adapters for Claude, Codex, Gemini, Ollama).
- Native Anthropic Model Context Protocol (MCP) server & client gateway.

03 // DATA & ISOLATION LAYER

137 Schema Tables & Git Worktrees

- Embedded PostgreSQL runtime: zero external Docker dependencies needed.
- 137 typed Drizzle ORM relational entity tables (`packages/db/src/schema/*.ts`).
- Isolated Git worktree execution workspaces per task (`services/execution-workspaces.ts`).
- AES-256-GCM encrypted secrets vault with per-agent credential scoping.

The Heartbeat Concurrency Engine & Turn Allocation Loop

Audited code evidence: server/src/services/heartbeat.ts (29,391 lines)

HEARTBEAT EXECUTION LIFECYCLE

Step 1 // Wake & Issue Checkout

Scheduler wakes on 15s intervals, queries active issues, checks company daily token budget, and claims available issues via PostgreSQL row locks.

Step 2 // Worktree & Workspace Provisioning

Clones a pristine Git worktree checkout on an isolated task branch, preventing unverified modifications to master.

Step 3 // ACP/ACPX Agent Dispatch

Initiates agent execution sidecar over Agent Client Protocol, streaming real-time prompt thoughts and tool invocations.

Step 4 // Tool Policy & Amber Gate Check

Inspects tool calls against security policies. High-risk actions halt execution pending human confirmation.

Step 5 // Turn Settlement & Lock Release

Records token consumption in costs ledger, settles execution status, and streams live events to WebSockets.

CONCURRENCY, TIMEOUTS & RESILIENCE

★ Orphaned Lock Settlement

If a server crashes or an agent hangs, the watchdog detects stale timestamps (>300s) and automatically settles locks without human intervention.

★ Watchdog Execution Timeouts

Enforces strict wall-clock execution deadlines per turn (e.g. 180s) to kill zombie subprocesses and prevent infinite recursion.

★ Database Row-Level Locking

Uses `SELECT ... FOR UPDATE SKIP LOCKED` semantics in PostgreSQL to enable horizontal multi-worker dispatch without race conditions.

★ Live Event Streaming Bus

All state transitions broadcast across WebSockets to Monaco editors, SVG org charts, and external chat channels simultaneously.

Model Context Protocol Gateway & Cryptographic Amber Approvals

Audited code evidence: server/src/services/tool-gateway.ts & decision-signing.ts

ANTHROPIC MCP TOOL GATEWAY

- **Native Model Context Protocol**
  Standardized on Anthropic's open MCP specification. Exposes database, filesystem, git, and web search tools via JSON-RPC stdio.
- **RBAC Tool Permission Matrix**
  Tools are partitioned across permission tiers. An intern agent cannot invoke production deployment tools; a QA agent cannot merge pull requests.
- **Automatic Secret Redaction**
  All tool arguments and returned stdio payloads pass through a regex and entropy sanitizer to scrub API keys and passwords.
- **Pluggable Custom MCP Servers**
  Enterprises can plug in custom internal MCP servers (Jira, Datadog, Snowflake) by adding a single config block in `companies.sh`.

AMBER GATES & HMAC DECISION SIGNING

- **★ Amber Gate Approval Engine**
  When an agent attempts a sensitive operation (e.g. `rm -rf`, `DROP TABLE`, `git push --force`), the turn pauses instantly. A notification appears in the web UI and Slack.
- **★ Interactive Operator Modal**
  Human operators can inspect full command diffs, approve, reject with feedback, or edit tool arguments in-flight (`server/src/routes/approvals.ts`).
- **★ HMAC-SHA256 Cryptographic Signing**
  Every approved decision generates an HMAC-SHA256 signature combining the task ID, operator ID, timestamp, and payload hash.
- **★ Tamper-Proof Audit Ledger**
  Appends immutable records to `packages/db/src/schema/decision_logs.ts`. Any post-facto modification invalidates the cryptographic signature.

Token Budget Enforcer, 5-Channel Chat Bridge & Pulse Self-Healing

Audited code evidence: server/src/services/budgets.ts, chat-channels.ts, pulse-service.ts

FINANCIAL TOKEN BUDGETING

USD Hard Daily Caps

- Real-time token metering across prompt, completion, and reasoning tokens.
- Per-company daily budget enforcement (`services/budgets.ts`).
- Automated circuit breaker pauses agent turns upon threshold saturation.
- Live USD spend visualization on dashboard.

MULTI-CHANNEL CHAT BRIDGES

Discord, Slack, Teams, TG

- 1.42MB multi-channel synchronization engine (`services/chat-channels.ts`).
- Bi-directional issue threading across Discord, Slack, Microsoft Teams, Telegram, and GitHub.
- Developers trigger agent tasks directly from Slack slash commands.

PULSE SELF-HEALING ENGINE

30s Automated Diagnostics

- Continuous 30-second automated health polling (`pulse-service.ts`).
- Validates database connectivity, AI model latency, and runner health.
- Automatic fallback model routing on HTTP 429 rate-limits.
- Zero manual intervention recovery.

Dynamic Git Worktree Sandboxing & Encrypted Secrets Vault

Audited code evidence: server/src/services/execution-workspaces.ts & secrets.ts

GIT WORKTREE TASK ISOLATION

- **Zero Repository Pollution**
  Instead of letting agents execute in dirty local directories, GeetorusAI dynamically spins up an isolated Git worktree for every single task.
- **Branch Checkout Automation**
  Automatically checks out isolated branch `agent/{task-id}`, executes code changes, runs local tests, and prepares clean git commits.
- **Pristine State Guarantee**
  If a task is rejected or fails verification, the entire worktree is destroyed, guaranteeing zero orphan files or broken builds on main.
- **Linux Bubblewrap & Docker Sandboxing**
  Supports hardware-level Linux user namespace sandboxing (Bubblewrap) and ephemeral Docker containers for dangerous execution.

ENCRYPTED SECRETS VAULT & AIR-GAPPED DEPLOY

- **★ AES-256-GCM Secret Encryption**
  Corporate API keys, SSH keys, and database credentials are stored in `company_secrets` encrypted with AES-256-GCM. Zero plain-text secrets.
- **★ Per-Agent Credential Scoping**
  Agents only receive decrypted environment variables for tools explicitly granted in their role profile at runtime.
- **★ 100% Air-Gapped / Zero-Telemetry Deployment**
  Regulated enterprises (defense, healthcare, banking) can run GeetorusAI entirely offline using local Ollama models (Llama 3.3, DeepSeek-R1).
- **★ Sovereign Data Storage**
  All databases, git branches, and logs reside strictly within customer infrastructure. Zero data exfiltration to proprietary SaaS clouds.

P0, P1, P2 Strategic Roadmap to Outperform All Competitors

Technical Specification: docs/pitch/feature_roadmap.md

PHASE P0 // IMMEDIATE MOAT

Months 1-3 (Post-Seed)

- Distributed BullMQ / Redis Task Queue (horizontal worker scaling).
- Vector Semantic Memory (pgvector embeddings for cross-session recall).
- Enterprise SSO & Fine-Grained RBAC (Okta / Azure AD / SAML).
- Automated Multi-Model Cost Router (Haiku 5.5 / o4-mini micro-routing).

PHASE P1 // ENTERPRISE EXPANSION

Months 4-7

- Automated Compliance Engine (SOC2 Type II, ISO 27001, DPDP export).
- Native Desktop Companion App (Tauri / Rust wrapping local CLIs).
- Automated Test Generation & Mutation Evals.
- Multi-Tenant Billing Gateway (Stripe/Paddle seat & token metering).

PHASE P2 // ECOSYSTEM LOCK-IN

Months 8-12

- Universal Agent & Tool Marketplace (one-click deployable teams).
- Cryptographic Zero-Knowledge Audit Attestation.
- Autonomous Agent Self-Optimization (DPO fine-tuning loop).
- Policy-as-Code Governance Kernel (Open Policy Agent / Rego).

PROPRIETARY NOTICE: Deep internal heuristics, fine-tuned agent prompt DAGs, and benchmark logs are deliberately withheld here. Available under NDA in the Investor Data Room.

Build the Sovereign Multi-Agent Future With Us

Self-Hosted, Audited, Governed AI Workforces for the Enterprise.

GEETORUSAI

The Autonomous AI Agent Control Plane & Virtual Company Operating System

- Founder: Udhaya Kumar (CEO & Founder, Geetorus)
- Engineering Hub: Tamil Nadu, India (20 Core Researchers & Engineers)
- Code Repository: https://github.com/geetorusai/geetorus
- Documentation & Data Room: Available upon request under NDA
- Seed Financing Ask: $1,500,000 USD for 15.0% Equity